

Statement of Purpose

Amazon ML Summer School 2026

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My journey into AI/ML began not in a classroom, but out of curiosity — a desire to understand how machines could learn from data and make meaningful decisions. Over the past two years, that curiosity has matured into hands-on expertise across classical ML, deep learning, NLP, and computer vision.

One of my most impactful projects was a **Twitter Sentiment Analysis system** aimed at helping businesses decode customer feedback at scale. I built a deep learning model capable of classifying text into Positive, Negative, and Neutral categories. The challenge wasn't just model accuracy — it was understanding the noisy, informal nature of social media language. Working through tokenization strategies, handling slang and abbreviations, and experimenting with LSTM and transformer-based architectures gave me a genuine appreciation for the complexity of NLP in real-world settings. The project reinforced how impactful ML can be when deployed on problems that directly affect business decisions.

My second major project tackled the growing threat of synthetic media: **DeepFake Detection using transfer learning**. I built a binary image classifier that distinguishes manipulated faces from authentic ones, leveraging MobileNetV2 as a frozen feature extractor with a fine-tuned classification head. Designing the model to run efficiently on CPU was a deliberate constraint — I wanted to build something practical and deployable, not just academically impressive. This project deepened my understanding of computer vision pipelines, transfer learning, and the ethical implications of AI-generated media.

Across both projects, I worked extensively with Python, TensorFlow/Keras, and standard data science libraries. I've completed coursework in machine learning, linear algebra, and probability, and regularly engage with research papers and online resources to stay current.

Yet I am acutely aware of my gaps. While I can build and fine-tune models, my theoretical understanding of optimization algorithms, probabilistic graphical models, and the mathematical foundations underpinning modern ML is something I want to deepen significantly. I also have limited exposure to large-scale ML systems, distributed training, and MLOps — areas where Amazon's practical expertise is unmatched.

The Amazon ML Summer School represents exactly the bridge I need. Learning directly from Amazon scientists — who operate at the intersection of research and production-scale deployment — would give me exposure to the rigour and systems thinking that I cannot replicate through self-study alone. Topics such as reinforcement learning, causal inference, and foundation model training are areas where structured, expert-led learning would have an outsized impact on my trajectory.

I bring technical depth, genuine intellectual curiosity, and a track record of building real projects that solve real problems. I am ready to go beyond tutorials and engage seriously with the material. I believe MLSS is the catalyst that will transform me from a capable practitioner into a principled ML engineer — and I am eager to make the most of every moment of it.